Discovering Latent Musical Structure Using Spotify Audio Features

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***Abstract*—Digital music platforms generate large volumes of audio data that describe the acoustic characteristics of songs. While genre labels and popularity metrics offer surface-level categorization, they often fail to capture deeper musical similarities. This project proposes a data mining approach to discover latent musical structures using Spotify audio features. By applying dimensionality reduction, clustering, and anomaly detection techniques, the project aims to uncover meaningful patterns in music and provide interpretable insights into how songs relate to one another beyond predefined labels.**

I. Introduction

The increasing availability of large-scale music datasets has created new opportunities for discovering patterns in musical content. Streaming services such as Spotify extract quantitative audio features that describe properties including rhythm, energy, and emotional tone. These features provide a rich foundation for exploratory analysis beyond traditional genre-based classification.

The objective of this project is to apply data mining techniques to identify latent structures in music using Spotify audio feature data. Rather than focusing on predictive accuracy, the project emphasizes pattern discovery and interpretation. By clustering songs based on acoustic similarity and identifying anomalous tracks, this study seeks to uncover interpretable musical groupings that reflect underlying sound characteristics.

II. Dataset Description

This project utilizes the Spotify Tracks Dataset, publicly available on Kaggle. The dataset contains over 110,000 tracks and includes both audio features and metadata for each song.

Key attributes include danceability, energy, loudness, speechiness, acousticness, instrumentalness, liveness, valence, tempo, and duration. Additional metadata such as popularity scores and explicit content indicators provide context for interpreting discovered patterns. The dataset primarily consists of numeric features, making it well-suited for clustering, dimensionality reduction, and anomaly detection techniques.

Some preprocessing will be required to normalize feature scales, handle missing values, and remove non-informative identifiers such as track names and artist IDs.

III. Discovery Questions

This project is guided by the following discovery-oriented questions:

- Are there natural clusters of songs based on their acoustic features such as energy, tempo, and acousticness?
- How do the discovered clusters differ in terms of popularity and explicit content, and what musical characteristics define each group?
- Are there anomalous tracks whose audio profiles significantly deviate from typical musical patterns, and what features contribute to their uniqueness?

These questions emphasize pattern discovery and interpretation rather than prediction.

IV. Planned Techniques

A. Preprocessing and Transformation

Data preprocessing will include handling missing values, normalizing numeric features, and removing irrelevant attributes. Principal Component Analysis (PCA) will be applied to reduce dimensionality and visualize latent structure within the data.

B. Clustering

Clustering algorithms such as K-Means and DBSCAN will be used to identify natural groupings of songs based on audio features. The results of different clustering methods will be compared to evaluate consistency and interpretability.

C. Anomaly Detection

Anomaly detection techniques will be applied to identify tracks that deviate significantly from common musical patterns. These anomalous songs will be examined to determine which features contribute most strongly to their distinctiveness.

D. Interpretation and Evaluation

Cluster validity metrics and visual analysis will be used to assess the quality of discovered patterns. Each cluster will be interpreted in musical terms to provide meaningful insights into the dataset.

V. Preliminary Timeline

- **Milestone 2:** Data preprocessing, exploratory analysis, and PCA visualization.
- **Milestone 3:** Application of clustering and anomaly detection techniques and comparative analysis.
- **Milestone 4:** Interpretation of results, validation of patterns, and preparation of the final report and visualizations.

Potential challenges include selecting appropriate clustering parameters and ensuring that discovered patterns are meaningful rather than spurious.

VI. Conclusion

This project proposes a data mining approach to uncover latent musical structures using Spotify audio features. By focusing on discovery rather than prediction, the study aims to reveal interpretable patterns that enhance understanding of musical similarity and diversity within large-scale music datasets.